Hayden Jones

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Education

San Jose State University San Jose, CA
Bachelor of Science in Computer Engineering *May 2026*

- Coursework and On-Campus Participation: Object-Oriented Programming, Advanced Algorithm Design, Circuit Analysis, Analog / Digital Circuit Design, Electronics for Computer Systems, Database System Design, Computer Networking, Cybersecurity, Compiler Design, Operating Systems; Software and Computer Engineering Society, BASE (NSBE Chapter), SJSU Robotics, SJSU Esports

Projects

32-bit RISC-V Processor | *Verilog, SystemVerilog* October 2025 - January 2026

- Architected a single-cycle 32-bit RISC-V processor in **Verilog**, implementing a custom ALU and a dual-decoder control unit to support base integers and hardware-accelerated multiplication/division for **RV32IM** support
- Designed modular RTL components including a 32-word register file, immediate generator, and distinct instruction/data memory modules to optimize the execution datapath
- Integrated an Object-Oriented **SystemVerilog** testbench utilizing interface bundling and assertions to continuously monitor datapath routing and architectural safety constraints

Drawalong | *C, Network Sockets* December 2025 - January 2026

- Developed a collaborative drawing application using **C** and SDL2 to enable real-time, multi-client canvas sharing over an internet connection
- Architected custom network packets utilizing the **sockets API** to ensure high-speed and reliable, lean transmission of drawing/canvas data between clients and the host
- Created solutions for cross-platform compatibility issues to ensure functionality across both **Linux** and **Windows**

Hayplayer - Custom MP3/FLAC Player | *KiCad, STM32, VLSI, C* February 2026 - Ongoing

- Architected a custom **mixed-signal PCB** to interface STM32 and dedicated audio decoder for high-fidelity playback
- Designed and iterated on advanced power management circuitry for load-sharing logic and power distribution
- Optimized component placement and trace characteristics to maintain signal integrity for both digital and analog lines

Experience

San Jose State University August 2022 – May 2025
Machine Learning Researcher *San Jose, CA*

- Previously: Conducted experiments to evaluate the comprehensibility of commercial machine translations using NLP techniques via **PyTorch** and **NumPy**
- Used results to create a presentation with solutions to improve translations for languages with less data than average as determined by metrics from Meta
- Explored the capabilities of State-Of-The-Art **Large Language Models** on explaining high school and college level mathematics such as Calculus in a controlled environment

Google May 2022 – August 2022
Software Engineering Intern *Palo Alto, CA*

- Built a system to train decision trees on medical data from the ground up in Python using **TensorFlow**
- Optimized model to yield up to a **19% increase** in accuracy over the current model
- Implemented pipeline to save and load configuration files for the training process using a tailored **Protocol Buffer** to ensure reproducible experiments for over **30 engineers** utilizing the project

Skills & Interests

Technical Skills: FPGA, Circuit Design, Soldering, Python, C, C++, Verilog, SystemVerilog, TCL, JavaScript, PyTorch, TensorFlow, FreeRTOS, AWS, Embedded Application Design, Machine Learning
Soft Skills: Communication, Analytical/Creative Problem Solving, Adaptability, Teamwork, Leadership, Work Ethic
Developer Tools: Git, Visual Studio Code, Neovim, KiCad, Intel Quartus II, AMD Vivado, LTSPICE